

Week 10: Word Representations

Deep Learning – IITM BS

Comprehensive Notes + Practice Problems

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1 Why Do We Need Word Representations?

1.1 One-Hot Vectors: The Naive Approach

For vocabulary size $|V|$, word i is represented as a vector of length $|V|$ with a 1 at position i and 0s elsewhere.

Problems:

- Very high dimensional ($|V|$ can be 50K–13M)
- Extremely sparse (only one non-zero entry)
- Euclidean distance between *any* two words $= \sqrt{2}$
- Cosine similarity between *any* two words $= 0$
- No semantic information: “cat” and “dog” are as different as “cat” and “airplane”

Goal: Learn dense, low-dimensional vectors where semantically similar words are close together.

2 Co-occurrence Matrix

2.1 Construction

Given a corpus and a window size k , the co-occurrence matrix X is a $|V| \times |V|$ matrix where $X[w, c]$ counts how many times word c appears within k positions of word w .

Building a Co-occurrence Matrix

Corpus (stop words removed):

1. “students read books library”
2. “teachers read books class”
3. “students teachers discuss library”

Window $k = 1$ (one word left, one word right).

Sentence 1: students–read, read–books, books–library

Sentence 2: teachers–read, read–books, books–class

Sentence 3: students–teachers, teachers–discuss, discuss–library

	stud.	read	books	library	teach.	class	discuss
students	0	1	0	0	1	0	0
read	1	0	2	0	1	0	0
books	0	2	0	1	0	1	0
library	0	0	1	0	0	0	1
teachers	1	1	0	0	0	0	1
class	0	0	1	0	0	0	0
discuss	0	0	0	1	1	0	0

Note: $X[\text{read}, \text{books}] = \mathbf{2}$ (appears in sentence 1 and sentence 2).

2.2 Issues with Raw Counts

- Stop words (“the”, “a”) have very high counts $\rightarrow$ dominate the matrix
- Matrix grows with vocabulary $\rightarrow$ very large and sparse
- Raw counts don’t normalize for word frequency

3 PMI (Pointwise Mutual Information)

PMI Formula – EXAM CRITICAL

$$\text{PMI}(w, c) = \ln \left(\frac{\text{count}(w, c) \times N}{\text{count}(w) \times \text{count}(c)} \right)$$

$\text{count}(w, c)$ = co-occurrence count of word w with context c
 $\text{count}(w)$ = total occurrences of w in the matrix (row sum)
 $\text{count}(c)$ = total occurrences of c in the matrix (column sum)
 N = sum of all entries in the co-occurrence matrix

PPMI (Positive PMI): $\text{PPMI}(w, c) = \max(0, \text{PMI}(w, c))$

Intuition: PMI measures whether two words co-occur *more than expected by chance*. If $\text{PMI} > 0$, they co-occur more than random. If $\text{PMI} < 0$, they co-occur less than random.

PMI Calculation Example

From the matrix above, compute $\text{PMI}(\text{read}, \text{books})$:

$\text{count}(\text{read}, \text{books}) = 2$

$\text{count}(\text{read}) = \text{row sum for read} = 1 + 0 + 2 + 0 + 1 + 0 + 0 = 4$

$\text{count}(\text{books}) = \text{col sum for books} = 0 + 2 + 0 + 1 + 0 + 1 + 0 = 4$

$N = \text{sum of entire matrix} = 2(1 + 1 + 2 + 1 + 1 + 1 + 1 + 1 + 1) = 2 \times 11 = 22$

Wait—let me recount. Each pair is counted symmetrically. Sum all entries:

Row sums: students=2, read=4, books=4, library=2, teachers=3, class=1, discuss=2. Total $N = 18$.

$\text{PMI}(\text{read}, \text{books}) = \ln \left(\frac{2 \times 18}{4 \times 4} \right) = \ln \left(\frac{36}{16} \right) = \ln(2.25) \approx 0.811$

PMI is positive $\Rightarrow$ “read” and “books” co-occur more than expected.

4 SVD for Dimensionality Reduction

Given PPMI matrix $X \in \mathbb{R}^{m \times n}$, SVD gives:

$$X \approx \hat{X} = U_{m \times k} \Sigma_{k \times k} V_{k \times n}^T$$

Word representation: $W_{\text{word}} = U \Sigma \in \mathbb{R}^{m \times k}$ (rows are word vectors)

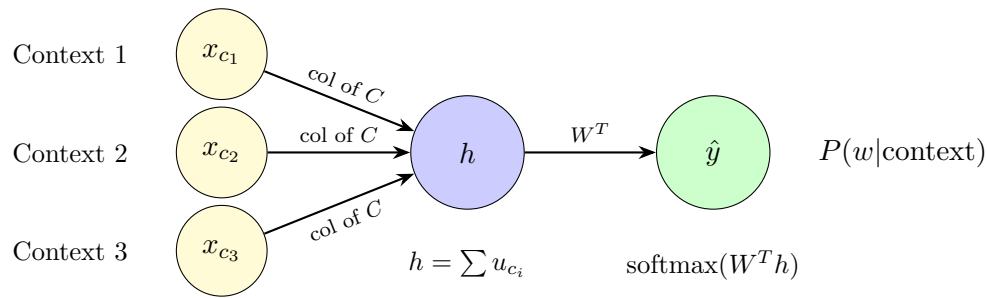
Context representation: $W_{\text{context}} = V \in \mathbb{R}^{n \times k}$

SVD Benefits

- Reduces high dimensionality to k dimensions
- Removes sparsity $\rightarrow$ dense embeddings
- Captures latent semantic similarity
- Best rank- k approximation (minimizes Frobenius norm error)

5 CBOW Model (Continuous Bag of Words)

Task: Given context words, predict the target (center) word.



CBOW Forward Pass – STEP BY STEP

Parameters: $C \in \mathbb{R}^{k \times |V|}$ (context matrix), $W \in \mathbb{R}^{|V| \times k}$ (word matrix)

Step 1: Get context embeddings. For one-hot input x_c (context word at index c):

$$u_c = C \cdot x_c = \text{column } c \text{ of } C$$

Step 2: If multiple context words, **sum** (not concatenate):

$$h = \sum_{\text{context words}} u_{c_i}$$

Step 3: Compute scores for every word in vocabulary:

$$z_j = w_j^T \cdot h \quad \text{where } w_j = \text{column } j \text{ of } W$$

(Equivalently: $z = W^T h$ or compute $W \cdot h$ depending on matrix convention)

Step 4: Apply softmax:

$$\hat{y}_j = P(w = j | \text{context}) = \frac{e^{z_j}}{\sum_{j'=1}^{|V|} e^{z_{j'}}}$$

Step 5: Loss:

$$L = -\ln(\hat{y}_{\text{true}}) = -\ln P(\text{correct word} | \text{context})$$

CBOW Worked Example

Vocabulary = {good(0), bad(1), ugly(2)}. Context weight matrix:

$$C = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -1 & -1 \end{bmatrix}, \quad W = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$$

Context window = 1. Compute $P(\text{bad} | \text{good})$.

Step 1: $u_{\text{good}} = C[:, 0] = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (column 0 of C)

Step 2: Only one context word, so $h = u_{\text{good}} = [1, 1]^T$

Step 3: Scores: $z = W^T h$. But W is 2×3 , so W^T is 3×2 :

$$z = W^T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

Step 4: Softmax: $e^0 = 1$, $e^1 = 2.718$, $e^{-1} = 0.368$. Sum = 4.086.

$$P(\text{bad} | \text{good}) = \frac{e^1}{4.086} = \frac{2.718}{4.086} \approx \mathbf{0.665}$$

5.1 Gradient Update for Word Vectors

CBOW Gradient

For the correct word w (true label) with context embedding u_c :

$$\frac{\partial L}{\partial v_w} = u_c \cdot (\hat{y}_w - 1) \quad (\text{correct word})$$

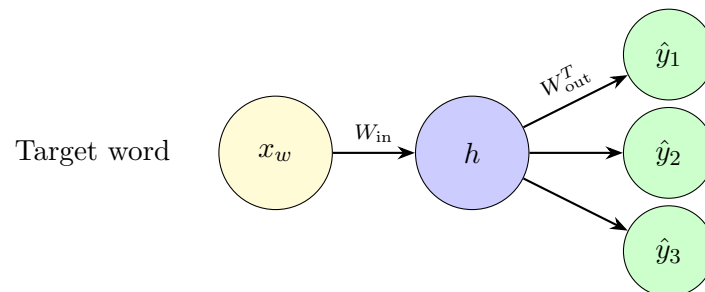
$$\frac{\partial L}{\partial v_{w'}} = u_c \cdot \hat{y}_{w'} \quad (\text{wrong words})$$

Update: $v_w^{\text{new}} = v_w - \eta \cdot \nabla v_w$

The training objective *increases cosine similarity* between word and its context.

6 Skip-Gram Model

Task: Given a target (center) word, predict the context words.



6.1 Counting Skip-Gram Pairs

Skip-Gram Pair Counting – EXAM FAVORITE

For a sentence of length L words, window size d (each side):

$$\text{Total pairs} = \sum_{i=0}^{L-1} \left[\min(d, i) + \min(d, L - 1 - i) \right]$$

Quick rule: Interior words contribute $2d$ pairs. Edge words contribute fewer.

Skip-Gram Counting Example

Sentence: “Learning from data helps build intelligent systems for society” (9 words), $d = 2$.

Index	Word	Pairs
0	Learning	$\min(2, 0) + \min(2, 8) = 0 + 2 = 2$
1	from	$\min(2, 1) + \min(2, 7) = 1 + 2 = 3$
2	data	$2 + 2 = 4$
3	helps	$2 + 2 = 4$
4	build	$2 + 2 = 4$
5	intelligent	$2 + 2 = 4$
6	systems	$2 + 2 = 4$
7	for	$2 + 1 = 3$
8	society	$2 + 0 = 2$

Total = $2 + 3 + 4 + 4 + 4 + 4 + 4 + 3 + 2 = 30$

7 Negative Sampling & Hierarchical Softmax

7.1 Negative Sampling

Full softmax over $|V|$ is expensive. Instead:

- Let D = set of correct (word, context) pairs from corpus
- Let D' = set of k randomly sampled *incorrect* pairs per correct pair
- Maximize:
$$\sum_{(w,c) \in D} \ln \sigma(v_c^T v_w) + \sum_{(w,r) \in D'} \ln \sigma(-v_r^T v_w)$$

7.2 Hierarchical Softmax

Build a binary tree with $|V|$ leaves. Each word has a unique path from root to leaf. Probability = product of binary decisions along the path. Cost: $O(\log |V|)$ instead of $O(|V|)$.

8 GloVe (Global Vectors)

Combines count-based (SVD) and prediction-based approaches:

$$\min_{v_i, v_j, b_i, b_j} \sum_{i,j} f(X_{ij}) \cdot (v_i^T v_j + b_i + b_j - \ln X_{ij})^2$$

where $f(X_{ij})$ is a weighting function that caps very frequent co-occurrences:

$$f(x) = \begin{cases} (x/x_{\max})^\alpha & \text{if } x < x_{\max} \\ 1 & \text{otherwise} \end{cases}$$

9 Evaluating Word Representations

- **Semantic relatedness:** Compare cosine similarity of word vectors with human judgments
- **Synonym detection:** Given a word + 4 candidates, pick the one with highest cosine similarity
- **Analogy tasks:** “king – man + woman $\approx$ queen”

10 Practice Problems

Problem 1: Co-occurrence Matrix [2 marks]

Corpus (stop words removed):

1. “cat sat mat” 2. “dog sat mat” 3. “cat dog play”

Window = 1. Construct the co-occurrence matrix and find $X[\text{sat}, \text{mat}]$.

Problem 2: PMI Calculation [3 marks]

Using the co-occurrence matrix from Problem 1, compute $\text{PMI}(\text{sat}, \text{cat})$. State whether it is positive, zero, or negative.

Problem 3: Skip-Gram Pairs [2 marks]

Skip-gram, window $d = 2$. Sentence: “deep learning helps build models” (5 words). How many unique (target, context) pairs?

Problem 4: Skip-Gram Pairs (Longer) [3 marks]

Skip-gram, window $d = 3$. Sentence: “the quick brown fox jumps over the lazy dog” (9 words). How many unique pairs?

Problem 5: CBOW Forward Pass [3 marks]

Vocabulary = {apple(0), banana(1), cherry(2)}. Matrices:

$$C = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}, \quad W = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Context = “apple” (one word). Compute $P(\text{banana}|\text{apple})$.

Problem 6: CBOW with Multiple Context Words [3 marks]

Same matrices as Problem 5. Context words = {apple, cherry} (window $d = 1$, target is banana). Compute $P(\text{banana}|\text{apple}, \text{cherry})$.

Problem 7: Cross-Entropy Loss [2 marks]

CBOW model predicts probability 0.15 for the correct target word. Compute the cross-entropy loss. (Use natural log.)

Problem 8: Gradient Update [3 marks]

After CBOW forward pass: $\hat{y}_w = 0.4$ for the correct word w . Context embedding $u_c = [3, -2]^T$. Current word vector $v_w = [1, 1]^T$. Learning rate $\eta = 0.05$. Compute v_w^{new} using: $v_w^{\text{new}} = v_w - \eta \cdot u_c(\hat{y}_w - 1)$.

Problem 9: SVD Benefits [2 marks, MCQ]

Which problems of the co-occurrence matrix does SVD address? (Select all correct.)

- A. High dimensional representation B. Sparse matrix C. Captures latent semantic similarity
D. Increases vocabulary size E. Produces dense embeddings

Problem 10: Conceptual [2 marks, MCQ]

In CBOW, when we have multiple context words, we:

- A. Concatenate their one-hot vectors B. Sum their embedding columns C. Average their embedding columns
D. Take the max of their embeddings

11 Solutions

Problem 1: Pairs (window=1): Sent1: cat–sat, sat–mat. Sent2: dog–sat, sat–mat. Sent3: cat–dog, dog–play.

	cat	sat	mat	dog	play
cat	0	1	0	1	0
sat	1	0	2	1	0
mat	0	2	0	0	0
dog	1	1	0	0	1
play	0	0	0	1	0

$X[\text{sat}, \text{mat}] = \mathbf{2}$.

Problem 2: $\text{count}(\text{sat}, \text{cat}) = 1$. $N = 2 + 4 + 2 + 3 + 1 = 12$.

$\text{count}(\text{sat}) = 4$, $\text{count}(\text{cat}) = 2$. $\text{PMI} = \ln(1 \times 12 / (4 \times 2)) = \ln(12/8) = \ln(1.5) \approx 0.405$. **Positive**.

Problem 3: 5 words, $d = 2$: $0 + 2 = 2$, $1 + 2 = 3$, $2 + 2 = 4$, $2 + 1 = 3$, $2 + 0 = 2$. Total = **14**.

Problem 4: 9 words, $d = 3$: $0 + 3 = 3$, $1 + 3 = 4$, $2 + 3 = 5$, $3 + 3 = 6$, $3 + 3 = 6$, $3 + 3 = 6$, $3 + 2 = 5$, $3 + 1 = 4$, $3 + 0 = 3$. Total = **42**.

Problem 5: $u_{\text{apple}} = C[:, 0] = [1, 2]^T$. $h = [1, 2]^T$.

$$z = W \cdot h = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$$

$e^{-1} = 0.368$, $e^2 = 7.389$, $e^{-1} = 0.368$. Sum = 8.125.

$P(\text{banana}|\text{apple}) = 7.389/8.125 = \mathbf{0.909}$.

Problem 6: $u_{\text{apple}} = [1, 2]^T$, $u_{\text{cherry}} = C[:, 2] = [-1, 0]^T$.

$$h = [1, 2]^T + [-1, 0]^T = [0, 2]^T. \quad z = W \cdot h = \begin{bmatrix} -2 \\ 2 \\ 0 \end{bmatrix}$$

$e^{-2} = 0.135$, $e^2 = 7.389$, $e^0 = 1$. Sum = 8.524.

$P(\text{banana}) = 7.389/8.524 = \mathbf{0.867}$.

Problem 7: $L = -\ln(0.15) = -(-1.897) = \mathbf{1.897}$.

Problem 8: $v_w^{\text{new}} = [1, 1]^T - 0.05 \cdot [3, -2]^T(0.4 - 1) = [1, 1]^T - 0.05 \cdot [3, -2]^T(-0.6) = [1, 1]^T + [0.09, -0.06]^T = \mathbf{[1.09, 0.94]^T}$.

Problem 9: B, C, E. (A is addressed indirectly by producing low-dim embeddings, but the standard correct answers from IITM papers are B, C, E.)

Problem 10: B. Sum the embedding columns (not concatenate).